



RFC: Dune Ops Observability

Comprehensive Roadmap and Release Cadence

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Target: dune-ops-observability-addon, dune-awakening-selfhost-docker (R2-R5)

Abstract

This RFC defines the comprehensive roadmap for the Dune Ops Observability addon ecosystem: 5 Core infrastructure releases (R1-R5) and 6 public addon releases (v0.2-v1.0) with explicit cross-track dependencies. The addon release plan consolidates 7 originally planned releases into 4: v0.4 (Game Activity & Combat), v0.5 (Economy & Resources), v0.6 (World & Assets), and v1.0 (SOC/OPS Operations Center). Each release maps to industry-standard metrics and uses an 8-type tagging convention.

Motivation

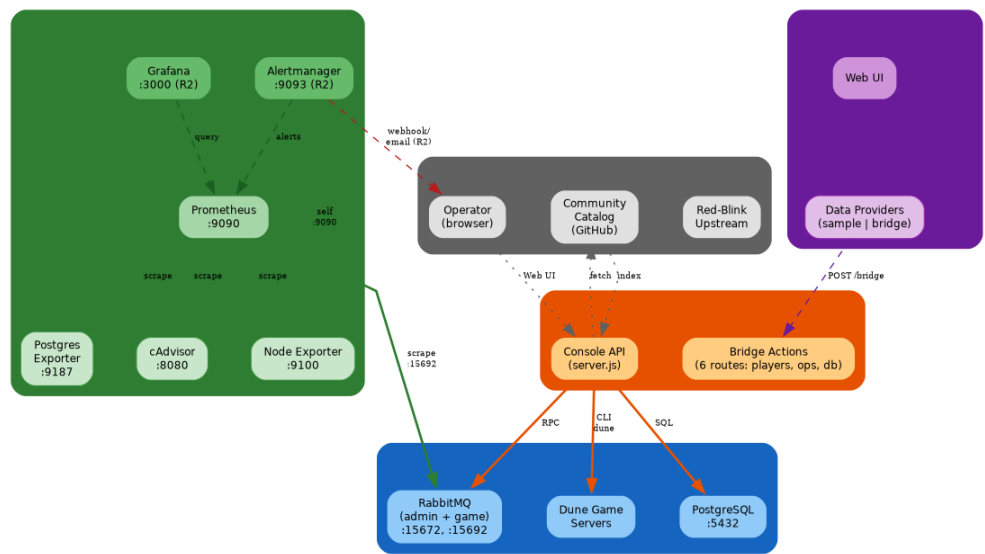
Critical gaps in the current system:

- No visualization — Prometheus data queryable only via raw PromQL; no Grafana
- No alerting — 16 alert rules have nowhere to route; no Alertmanager
- No game telemetry — dune-stack.yml is empty; no player/combat/economy metrics
- No Prometheus bridge — addon cannot display Prometheus metrics
- Rate limits in-memory — process restart clears all abuse buckets
- No proxy-aware IP — X-Forwarded-For not supported

Goals

- Close all gaps within 4 Core releases (R2-R5)
- Deliver game telemetry across 3 addon releases (v0.4-v0.6)
- Graduate to SOC/OPS center at v1.0 after Core R3+R4
- Maintain formal 5-gate governance with SOC2-style evidence

Current Architecture (R1)

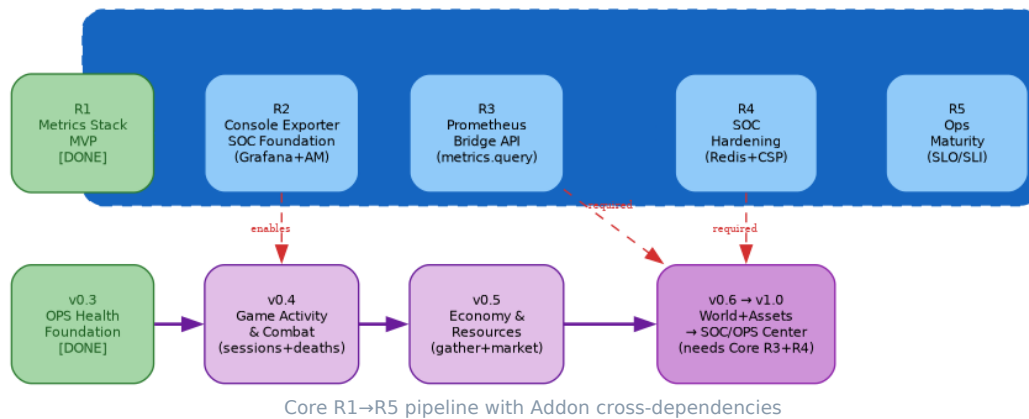


System architecture: Game Stack, Metrics Stack, Console API, Addon

Component	Container	Port	Access
Prometheus	dune-prometheus	127.0.0.1:9090	Local only
cAdvisor	dune-cadvisor	8080	Internal
Node Exporter	dune-node-exporter	9100	Internal
Postgres Exporter	dune-postgres-exporter	9187	Internal
RabbitMQ	dune-rmq-admin/game	15692	Internal

16 alerts across 4 groups. 7 bridge actions. SOC controls: HMAC sessions, CSRF, login/bridge rate limits, JSONL audit, addon provenance tracking.

Proposed Architecture



Core R2: Console Exporter + SOC Foundation

- Grafana (:3000) + Alertmanager (:9093) — both localhost, Grafana always-on
- dune-stack.yml populated (Console API RED metrics)
- Alertmanager: email+webhook

Core R3: Prometheus Bridge API

- metrics.query bridge action (safe PromQL subset, ops:read)
- /api/server/metrics endpoint (host CPU/mem/disk)
- Structured logging (JSON Lines)

Core R4: SOC Hardening

- Redis for persistent rate limits (survives restart)
- X-Forwarded-For + ADMIN_TRUSTED_PROXIES
- Per-addon CSP sandbox

Core R5: Ops Maturity

- SLO/SLI framework (99.5% uptime, p95<5s, freshness<5min)
- Prometheus retention/backup, audit log rotation
- DORA tracking (deploy freq, MTTR, change fail%)

Addon Release Train (v0.4-v1.0)

v0.4.0 — Game Activity & Combat (merges v0.4+v0.5)

Metrics: Sessions, transitions, retention, deaths by location/cause, PvP/PvE, NPC kills, death spikes

Required: R2 (Grafana+Alertmanager)

v0.5.0 — Economy & Resources (merges v0.6+v0.7)

Metrics: Ore/spice/fiber gathering, currency flow, market volume, inflation, scarcity indicators

Required: database discovery

v0.6.0 — World & Assets (merges v0.8+v0.9)

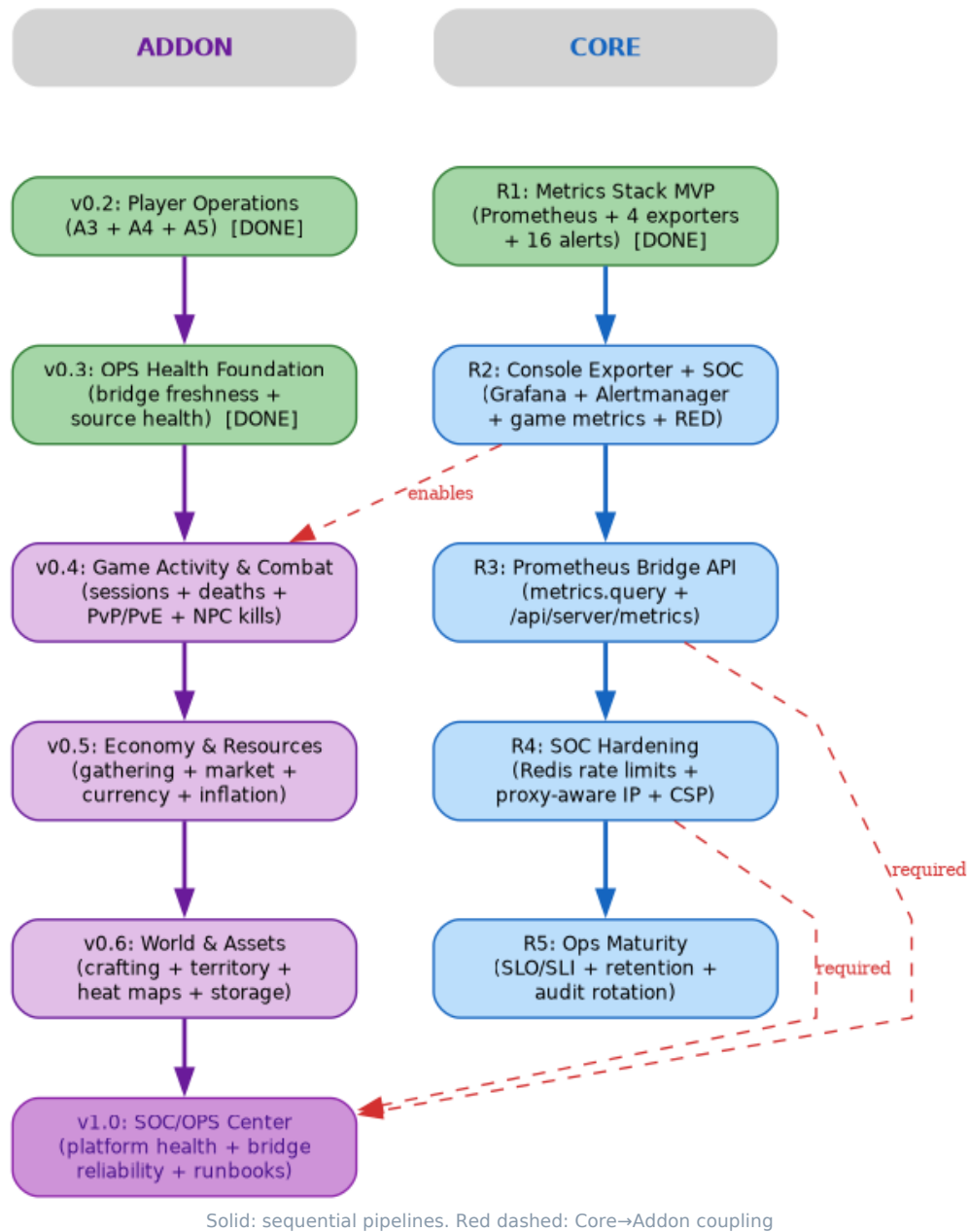
Metrics: Crafting volumes, storage pressure, territory hot spots, activity heat maps, base activity

Required: database discovery

v1.0.0 — SOC/OPS Center (merges v1.0)

Metrics: Platform health, bridge reliability, Prometheus display, addon drift, runbooks

Required: Core R3+R4



Tagging Convention

Context	Pattern	Example	Repo
Core feature merges	core-{domain}-r{n}	core-metrics-r2	Core fork
Addon releases	v{major}.{minor}.{patch}	v0.4.0	Addon
Addon RCs	v{major}.{minor}.{patch}-rc{n}	v0.4.0-rc1	Addon
Addon features	feature/{area}	feature/player-activity	Addon
Immutable snapshots	preserve/{desc}	preserve/pre-db-discovery	Addon

Backwards Compatibility

All existing bridge actions, CLI commands, manifest schema, and versioning remain unchanged. R2-R5 are additive only. No breaking changes planned.

Security Considerations

Layer 1: Transport
HMAC-SHA256 sessions
CSRF tokens on mutations
Security headers (nosniff, DENY)
timingSafeEqual password compare

Layer 2: Rate Limiting
Login: per-client 8/15m + global 32/15m
Bridge: per-addon 60/min + global 300/min
R4: Redis-backed persistence + proxy-aware IP

Layer 3: Input Validation
JSON body size limit (2MiB)
Upload size limit (1GiB)
IP allowlisting (ADMIN_ALLOWED_IPS)
Addon ID pattern validation (`^[a-z0-9-]{1,63}$`)

Layer 4: Bridge Access Control
Permission gating (ops:read, players:read)
Read-only SQL enforcement
SHA-256 checksum verification
Addon provenance tracking

Layer 5: Data Protection
Secret redaction (JWT, tokens, passwords)
JSONL audit logging
R4: Per-addon CSP sandbox
R4: Proxy-aware IP detection

Layer 6: CI/CD Security
Gitleaks secret scanning
Semgrep static analysis
Trivy filesystem vulnerability scan
ShellCheck + git diff --check

Key rules: Grafana+Alertmanager bind localhost. Redis on internal net with optional AUTH. metrics.query sanitizes PromQL (no unbounded cardinality). X-Forwarded-For only parsed behind trusted proxy CIDR. Per-addon CSP computed from declared permissions. All releases follow non-negotiable security rules from release-standard.md.

Open Issues

- PostgreSQL direct access vs bridge-mediated queries for game telemetry
- Grafana dashboard provisioning (JSON shipped vs user-built)
- Alertmanager channels beyond email/webhook (SMS/PagerDuty)
- Persistent audit log queryability through addon
- Game telemetry retention policy
- Core upstream PR scope — which R2-R5 components go to Red-Blink

This RFC is open for review via PR or issue on [yacketrj/dune-ops-observability-addon](#). Last updated: 04 July 2026.